

Matplotlib is a popular Python library used for:

Data visualization Creating graphs and charts

**It is widely used in:**

Data Science

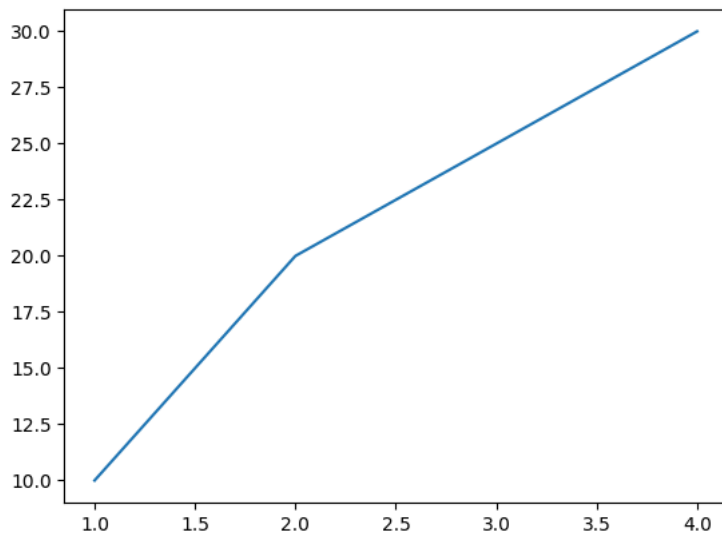
Machine Learning

Research work

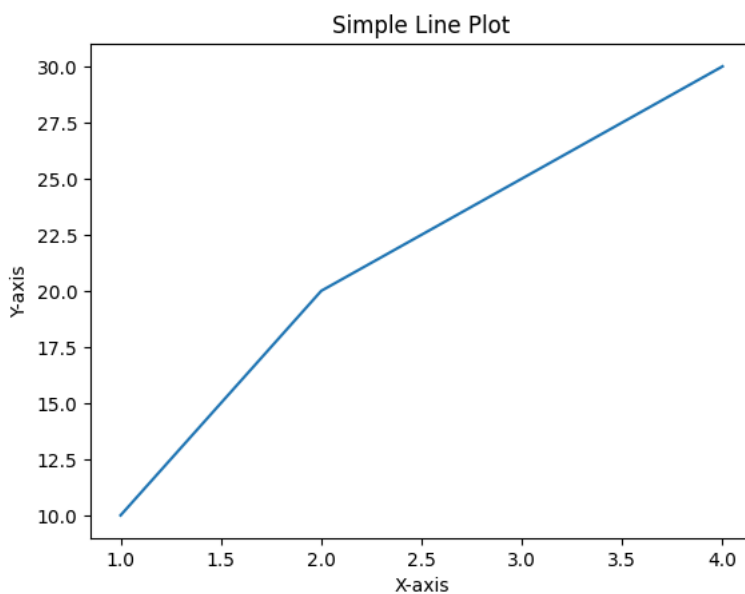
```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4]  
y = [10, 20, 25, 30]
```

```
plt.plot(x, y)  
plt.show()
```



```
plt.plot(x, y)  
plt.xlabel("X-axis")  
plt.ylabel("Y-axis")  
plt.title("Simple Line Plot")  
plt.show()
```



### Different Types of Plots

1. Line Plot
2. Bar Plot

### 3. Scatter Plot

### 4. Histogram

#### Line Plot

To show trends over time or continuous data

Example:

Temperature changes over day

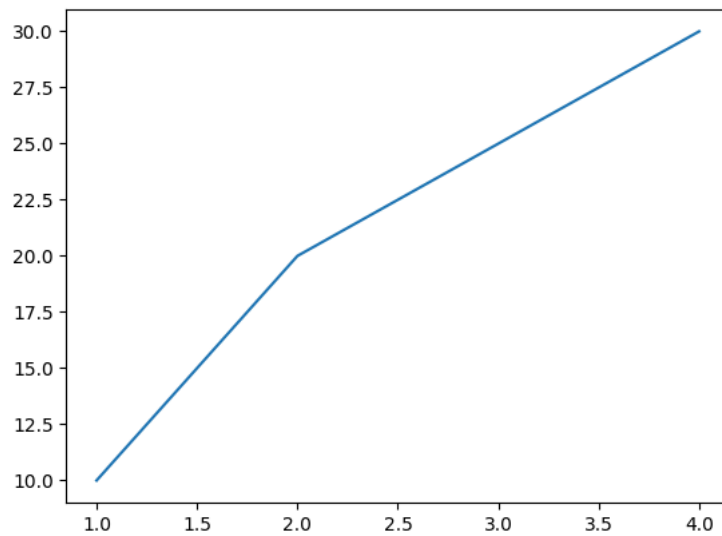
Stock prices over months

Why use it?

Helps you see increase, decrease, patterns, trends

```
plt.plot(x, y)
```

```
[<matplotlib.lines.Line2D at 0x78374d10b650>]
```



#### Bar Plot

To compare different categories

Example:

Marks of students in different subjects

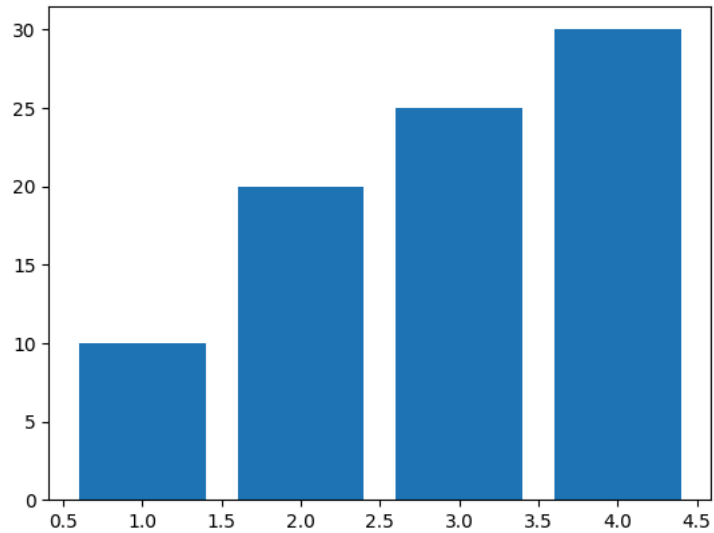
Sales of different products

Why use it?

Easy to compare values across groups

```
plt.bar(x, y)
```

<BarContainer object of 4 artists>



### Scatter Plot

To show relationship between two variables

Example:

Study hours vs marks

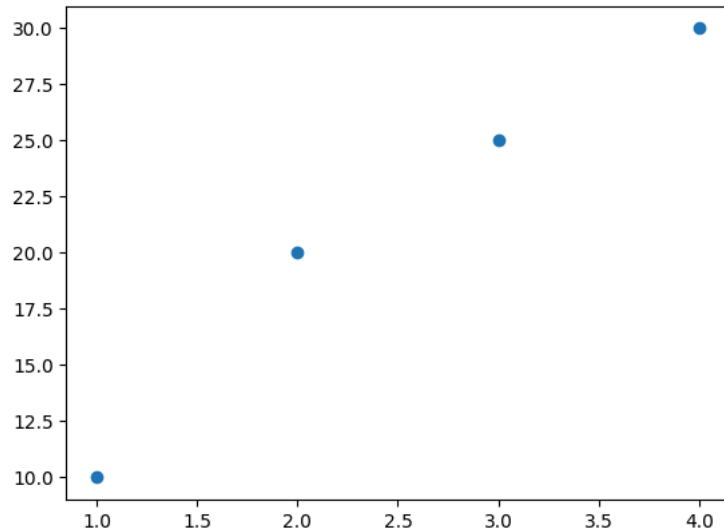
Height vs weight

Helps identify:

Correlation (positive/negative), Outliers

```
plt.scatter(x, y)
```

<matplotlib.collections.PathCollection at 0x78374dab7c20>



### Histogram

To show the distribution of data

Example:

Distribution of students' marks

Age distribution in a population

Shows:

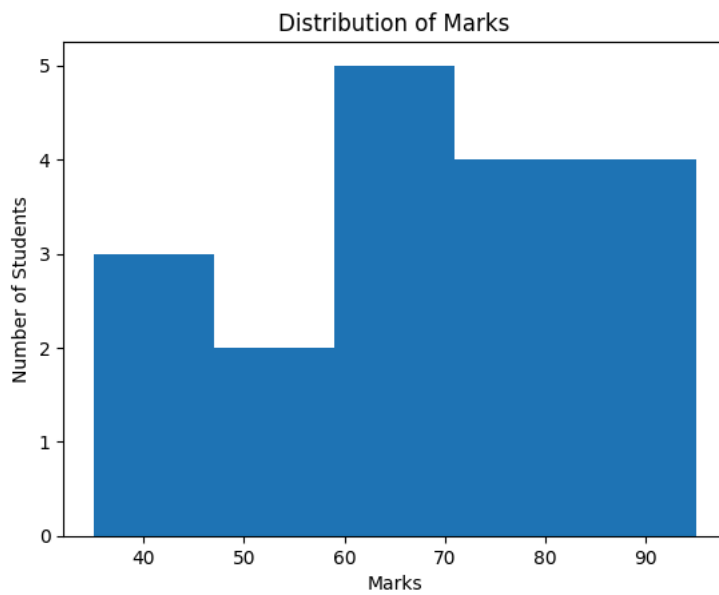
Frequency

```
import matplotlib.pyplot as plt
```

```
marks = [35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 60, 70, 75, 80, 85]
```

```
plt.hist(marks, bins=5) # divide marks into 5 groups (intervals)

plt.title("Distribution of Marks")
plt.xlabel("Marks")
plt.ylabel("Number of Students")
```



### Making Subplots

Subplots allow multiple graphs in one figure

Syntax:

```
plt.subplot(rows, columns, position)
```

```
x = [1, 2, 3, 4]
y1 = [10, 20, 25, 30]
y2 = [30, 25, 20, 10]

# First plot
plt.subplot(1, 2, 1)
plt.plot(x, y1)
plt.title("Plot 1")

# Second plot
plt.subplot(1, 2, 2)
plt.plot(x, y2)
plt.title("Plot 2")

plt.show()
```

